**BM3122 | MEDICAL IMAGING**

DOCUMENT B | MODEL 1

# Best-fit predicted practice paper

University of Moratuwa-style undergraduate practice examination. Independently prepared; this is not an official university examination or a disclosure of the future paper.

## BM3122 MEDICAL IMAGING

### Time allowed: Two hours | Total: 100 marks

• This model has four questions. All questions carry 25 marks. Answer ALL questions.

• Use the following information sheet. This is a closed-book practice examination apart from the supplied sheet.

• Show numerical working, units and stated assumptions. Derive results only when the question explicitly asks for a derivation.

• All constants and numerical conditions required are supplied. If interpreting a qualitative question, state your interpretation clearly.

• Suggested timing: 30 minutes per main question. Use separate answer sheets. Marking schemes are proposed and appear only in the solution documents.

**Candidate: _____ Date: _____**

Practice design: Recent-pattern best fit.

# Information sheet | Model paper 1

This sheet is provided for these independent practice papers. Equations are corrected and assumptions are stated here; provision of a similar sheet in 2026 is not verified. Symbols have their usual meaning and units must be consistent.

| Area | Relationships / constants |
|---|---|
| Constants | $\gamma=267.52\times10^{6}$ rad $s^{-1}$ $T^{-1}$; $\gamma bar=\gamma/(2\pi)$. $h=6.63\times10^{-34}$ J s; $k=1.38\times10^{-23}$ J/K. Kelvin=°C+273.15. |
| Resolution and noise | FWHM of a Gaussian=$2\sqrt{(2\ln2)}\sigma$. I=O*h. $CNR=(S_A-S_B)/\sigma_N$. For independent noise $SNR_N=SNR_1\sqrt{N}$. ADC step=voltage span/$2^n$ for equal-width bins. |
| X-ray / CT | $I=I_0\exp(-\Sigma\mu_i x_i)$; $HVL=\ln2/\mu$. $p=-\ln(I/I_0)$. $HU=1000(\mu-\mu_{water})/\mu_{water}$. Water=0 HU; air≈-1000 HU. Effective focal spot f=F sinθ; coverage=2D tanθ. |
| Nuclear medicine | $A=-dN/dt=\lambda N$; $\lambda=\ln2/T_{half}$; $A(t)=A_0\exp(-\lambda t)$. $1/T_{eff}=1/T_{phys}+1/T_{bio}$ for independent exponential clearance. Non-paralyzable dead time: τ=(N-n)/(nN), where N,n are true/measured rates. |
| MRI encoding | $f_0=\gamma B_0/(2\pi)$. Full RF bandwidth $\Delta f=\gamma bar\,|G|\Delta z$. A central slice has $f_{min,max}=f_0\pm\Delta f/2$. $T_{scan}=TR\times N_{phase}\times NEX$ for conventional 2D acquisition. FOV=1/Δk. |
| MRI signal | Spoiled steady-state GE: $S=K\rho[(1-E_1)\sin\alpha/(1-E_1\cos\alpha)]\exp(-TE/T_2^*)$, $E_1=\exp(-TR/T_1)$. Ernst angle=$\arccos(E_1)$. Ideal spin echo: $S=K\rho[1-\exp(-TR/T_1)]\exp(-TE/T_2)$. Net population fraction=$\gamma hB_0/(4\pi kT)$. |
| Ultrasound | $Z=\rho c$. Intensity reflectance at normal incidence=$[(Z_2-Z_1)/(Z_2+Z_1)]^2$. λ=c/f. Round-trip propagation loss=2αfd, with d in cm, f in MHz and α in dB/(cm·MHz). Echo time=2d/c; frame rate≤$c/(2dN_{lines}N_{firings/line})$. Axial resolution=$N_{cycles}\lambda/2$; focal lateral resolution≈λF/L. |
| Optical / thermal / dose | Gaussian OCT axial resolution in air $\delta z=(2\ln2/\pi)\lambda_0^2/\Delta\lambda$; tissue $\delta z=\delta z_{air}/n$. Total thermal emission $M=\varepsilon\sigma T^4$. D=energy/mass (Gy); $H=w_R D$ (Sv); $E=\Sigma w_T H_T$ (Sv). Intensity loss from reflectance R: -10 $\log_{10}R$ dB. |

ART tasks use unit ray weights and additive residual / number-of-pixels updates as explicitly directed. Ultrasound timing assumes sequential lines unless stated otherwise. All models are answer-all papers. Useful approximate conversion: 1 mT/cm=0.1 T/m.

# MODEL 1 | Question 1 [25 marks]

## Image quality and nuclear medicine

**(a)** Define spatial resolution. State what the point spread function (PSF) represents and write the image-formation relationship for a linear, shift-invariant system. **[3]**

**(b)** A line spread function is L(x)=C exp[-(x-x0)^2/(2σ^2)]. Derive its full width at half maximum (FWHM). Evaluate the width for σ=0.60 mm. **[4]**

**(c)** Sketch the result of convolving (i) an ideal point and (ii) a narrow rectangular object with a normalized Gaussian PSF. Explain the effect on fine structures. **[3]**

**(d)** State four desirable properties of a SPECT radiotracer, explaining the imaging reason for each. **[4]**

**(e)** Using a labeled sketch, explain positron annihilation coincidence detection in PET. Give two benefits of combining PET with CT. **[5]**

**(f)** At t=0, two patients have in-body activities AA=240 MBq and AB=80 MBq. A has physical and biological half-lives of 6 h and 6 h. B has physical and biological half-lives of 12 h and 12 h. Assuming independent exponential clearance, find when their in-body activities are equal and the common activity. **[6]**

Total for Question 1: 25 marks

# MODEL 1 | Question 2 [25 marks]

## Projection radiography and computed tomography

**(a)** Draw a planar X-ray radiography system, labeling the tube, beam collimator, patient, anti-scatter grid and detector. State the distinct functions of collimator and grid. **[4]**

**(b)** State three consequences or trade-offs of reducing the effective focal spot. Include an image-quality consequence. **[3]**

**(c)** A monoenergetic beam illuminates 0.60 cm thick bars with HVL=0.20 cm. Gaps have negligible attenuation. Calculate μ, Ibar/Igap, and Weber contrast C=(Igap-Ibar)/Igap. State one reason this simple model is imperfect for a clinical polychromatic beam. **[5]**

**(d)** Give three advantages of CT over planar projection radiography. **[3]**

**(e)** State the central section theorem, with a labeled diagram relating projection space to the object Fourier domain. **[4]**

**(f)** A 2×2 image has unit-weight ray sums: top row 6, bottom row 10, left column 7 and right column 9. Start all pixels at zero. Perform one ART sweep in that order, adding each ray residual equally to its two pixels. Report every intermediate image and determine whether these data uniquely identify all four pixels. **[6]**

Total for Question 2: 25 marks

# MODEL 1 | Question 3 [25 marks]

## Magnetic resonance imaging

**(a)** Name three intrinsic tissue parameters that affect MRI pixel intensity and explain why intensity cannot be interpreted as proton density alone. **[3]**

**(b)** A 1.5 T scanner uses Gz=20 mT/m to select a 4.0 mm thick axial slice centred at z=0. Calculate the central Larmor frequency, full RF excitation bandwidth and frequency endpoints in Hz. **[4]**

**(c)** Sketch a basic gradient-echo imaging sequence on RF, Gslice, Gphase, Gread and ADC lines. Identify TE and explain the purpose of each gradient. **[5]**

**(d)** For one average, 192 phase-encoding steps and TR=0.80 s, calculate acquisition time. Each slice occupies a complete 20 ms sequence slot. Ignoring overhead and cross-talk, find how many slices can be interleaved without increasing that time. Explain how the selected slice is changed. **[4]**

**(e)** A spoiled steady-state GE scan has TR=0.50 T1. Find the Ernst angle. For fixed TR, TE and receiver noise, calculate the percentage SNR reduction when a 90° pulse replaces the optimum angle. **[4]**

**(f)** Explain why a conventional spin echo is less sensitive than GE to static magnetic-field inhomogeneity. Sketch the RF and signal timing and state which transverse relaxation time governs the ideal echo envelope. **[5]**

Total for Question 3: 25 marks

# MODEL 1 | Question 4 [25 marks]

## Ultrasound and optical coherence tomography

**(a)** Explain why an air gap between the ultrasound probe and skin is undesirable and how coupling gel helps. **[3]**

**(b)** Explain ultrasound speckle and give two acquisition methods for reducing it. **[3]**

**(c)** A small target lies 15 cm deep. Tissue attenuation is 0.80 dB/(cm·MHz). At most 80 dB propagation loss is allowed. Choose among 2, 3 and 5 MHz, ignoring reflection losses. Calculate the loss at each frequency and justify the choice for best resolution. Use c=1540 m/s. **[5]**

**(d)** A B-mode array has 192 elements fired in non-overlapping groups of six, one line per group. At 15 cm depth and c=1540 m/s, find the scan lines per frame and maximum frame rate. Assume one pulse per line, sequential lines and no overhead. **[4]**

**(e)** Describe the physical cause and image appearance of (i) reverberation and (ii) posterior acoustic enhancement. **[4]**

**(f)** Draw a basic time-domain OCT interferometer. Explain depth selection and the formation of A- and B-scans. State one medical application and one limitation. **[6]**

Total for Question 4: 25 marks

## End of Model Paper 1

DOCUMENT B | MODEL 2

# Alternative predicted practice paper

University of Moratuwa-style undergraduate practice examination. Independently prepared; this is not an official university examination or a disclosure of the future paper.

## BM3122 MEDICAL IMAGING

### Time allowed: Two hours | Total: 100 marks

• This model has four questions. All questions carry 25 marks. Answer ALL questions.

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• Show numerical working, units and stated assumptions. Derive results only when the question explicitly asks for a derivation.

• All constants and numerical conditions required are supplied. If interpreting a qualitative question, state your interpretation clearly.

• Suggested timing: 30 minutes per main question. Use separate answer sheets. Marking schemes are proposed and appear only in the solution documents.

**Candidate: _____ Date: _____**

Practice design: Alternative historical topic rotation.

# Information sheet | Model paper 2

This sheet is provided for these independent practice papers. Equations are corrected and assumptions are stated here; provision of a similar sheet in 2026 is not verified. Symbols have their usual meaning and units must be consistent.

| Area | Relationships / constants |
|---|---|
| Constants | $\gamma=267.52\times10^{6}$ rad $s^{-1}$ $T^{-1}$; $\gamma bar=\gamma/(2\pi)$. $h=6.63\times10^{-34}$ J s; $k=1.38\times10^{-23}$ J/K. Kelvin=°C+273.15. |
| Resolution and noise | FWHM of a Gaussian=$2\sqrt{(2\ln2)}\sigma$. I=O*h. $CNR=(S_A-S_B)/\sigma_N$. For independent noise $SNR_N=SNR_1\sqrt{N}$. ADC step=voltage span/$2^n$ for equal-width bins. |
| X-ray / CT | $I=I_0\exp(-\Sigma\mu_i x_i)$; $HVL=\ln2/\mu$. $p=-\ln(I/I_0)$. $HU=1000(\mu-\mu_{water})/\mu_{water}$. Water=0 HU; air≈-1000 HU. Effective focal spot f=F sinθ; coverage=2D tanθ. |
| Nuclear medicine | $A=-dN/dt=\lambda N$; $\lambda=\ln2/T_{half}$; $A(t)=A_0\exp(-\lambda t)$. $1/T_{eff}=1/T_{phys}+1/T_{bio}$ for independent exponential clearance. Non-paralyzable dead time: τ=(N-n)/(nN), where N,n are true/measured rates. |
| MRI encoding | $f_0=\gamma B_0/(2\pi)$. Full RF bandwidth $\Delta f=\gamma bar\,|G|\Delta z$. A central slice has $f_{min,max}=f_0\pm\Delta f/2$. $T_{scan}=TR\times N_{phase}\times NEX$ for conventional 2D acquisition. FOV=1/Δk. |
| MRI signal | Spoiled steady-state GE: $S=K\rho[(1-E_1)\sin\alpha/(1-E_1\cos\alpha)]\exp(-TE/T_2^*)$, $E_1=\exp(-TR/T_1)$. Ernst angle=$\arccos(E_1)$. Ideal spin echo: $S=K\rho[1-\exp(-TR/T_1)]\exp(-TE/T_2)$. Net population fraction=$\gamma hB_0/(4\pi kT)$. |
| Ultrasound | $Z=\rho c$. Intensity reflectance at normal incidence=$[(Z_2-Z_1)/(Z_2+Z_1)]^2$. λ=c/f. Round-trip propagation loss=2αfd, with d in cm, f in MHz and α in dB/(cm·MHz). Echo time=2d/c; frame rate≤$c/(2dN_{lines}N_{firings/line})$. Axial resolution=$N_{cycles}\lambda/2$; focal lateral resolution≈λF/L. |
| Optical / thermal / dose | Gaussian OCT axial resolution in air $\delta z=(2\ln2/\pi)\lambda_0^2/\Delta\lambda$; tissue $\delta z=\delta z_{air}/n$. Total thermal emission $M=\varepsilon\sigma T^4$. D=energy/mass (Gy); $H=w_R D$ (Sv); $E=\Sigma w_T H_T$ (Sv). Intensity loss from reflectance R: -10 $\log_{10}R$ dB. |

ART tasks use unit ray weights and additive residual / number-of-pixels updates as explicitly directed. Ultrasound timing assumes sequential lines unless stated otherwise. All models are answer-all papers. Useful approximate conversion: 1 mT/cm=0.1 T/m.

# MODEL 2 | Question 1 [25 marks]

## Image acquisition and radiotracer kinetics

**(a)** Name three modalities using ionizing radiation and three non-ionizing modalities. For this answer, treat PET and SPECT as distinct modalities. **[3]**

**(b)** For CT, MRI, B-mode ultrasound and PET, state the principal physical quantity or signal determining pixel brightness. Include one qualification for MRI or ultrasound. **[4]**

**(c)** Define quantization error and explain two ways to reduce its relative importance in image acquisition. **[3]**

**(d)** A laboratory needs 18 MBq at 10:00 tomorrow. A sealed tracer stock with physical half-life 4 h is purchased at 18:00 today. Neglect biological clearance. Find the required purchase activity. **[4]**

**(e)** Initially there are equal numbers of nuclei of tracers X and Y. Their half-lives are 4 h and 20 h. With equal and constant count-detection efficiencies and no biological clearance, find the earliest time at which X contributes only 25% of their combined count rate. **[5]**

**(f)** Compare SPECT and PET in terms of emission, direction selection and detection sensitivity. Give one practical limitation of PET. **[6]**

Total for Question 1: 25 marks

# MODEL 2 | Question 2 [25 marks]

## Digital subtraction angiography and CT geometry

**(a)** For a ray through static tissue thickness xt and vessel thickness xv, write pre-contrast intensity I1. Use incident intensity I0 and attenuation coefficients μt and μb; assume monoenergetic radiation and negligible scatter. **[2]**

**(b)** After contrast injection, the vessel coefficient changes from μb to μc while other conditions are unchanged. Write I2. **[2]**

**(c)** Show why I2-I1 does not eliminate the static tissue signal. **[3]**

**(d)** Show that ln(I2)-ln(I1) removes the static tissue term. Evaluate it for μc=1.8 cm^-1, μb=0.2 cm^-1 and xv=0.30 cm. State one acquisition condition required for cancellation. **[4]**

**(e)** An object has attenuation μ0=0.25 cm^-1 inside a disk of radius R=2.0 cm and zero outside. Derive its parallel-ray projection p(s,θ), where s is the signed perpendicular distance from the disk centre. Evaluate p at s=1.2 cm. **[4]**

**(f)** Compare third- and fourth-generation CT scanner geometry. Give three distinct features of each, supported by simple labeled sketches. **[6]**

**(g)** Name four routine CT quality-assurance tests and state what each tests. **[4]**

Total for Question 2: 25 marks

# MODEL 2 | Question 3 [25 marks]

## MRI hardware and phantom measurements

**(a)** Draw the main components of an MRI scanner and explain the functions of the main magnet, three gradient-coil sets, and RF transmit/receive coils. **[6]**

**(b)** State three requirements of a main imaging magnet and explain why each matters. **[3]**

**(c)** Using the provided high-temperature spin-population approximation, calculate the net proton population imbalance (Nparallel-Nantiparallel)/Ntotal as a percentage at B0=3.0 T and 27°C. Explain why the phantom volume is unnecessary. **[4]**

**(d)** A central 5.0 mm thick slice is selected with Gz=0.40 mT/cm at 3.0 T. Calculate the full RF bandwidth and absolute frequency endpoints. **[4]**

**(e)** A conventional single-slice acquisition has 240 readout samples and 160 phase-encoding steps. TR=1.25 s and two averages are acquired. Calculate the scan time. If only readout samples are doubled, explain the effect on the ideal scan time under the same TR. **[4]**

**(f)** Give two phantom-based MRI checks and explain how an appropriate phantom feature would support each measurement. **[4]**

Total for Question 3: 25 marks

# MODEL 2 | Question 4 [25 marks]

## Ultrasound and thermal imaging

**(a)** State three advantages and three limitations of ultrasound compared with CT or MRI. **[3]**

**(b)** Explain reflection, scattering and absorption of ultrasound in tissue. Link each to an image consequence. **[6]**

**(c)** A target lies 18 cm deep in tissue with attenuation 0.70 dB/(cm·MHz). The receiver accepts at most 95 dB propagation loss. Choose the highest usable frequency from 2, 3 and 4 MHz. Justify using calculated losses and the resolution trade-off. **[5]**

**(d)** At 18 cm imaging depth and c=1500 m/s, the required B-mode frame rate is 50 frames/s. Find the maximum integer number of sequential scan lines per frame, ignoring overhead. **[3]**

**(e)** With a labeled block diagram, explain how an uncooled thermal camera forms a medical thermogram. **[4]**

**(f)** State two appropriate medical uses of thermal imaging and link each to the physiological information observed. **[2]**

**(g)** Give two advantages of thermal imaging and two acquisition controls needed for reliable comparisons over time. **[2]**

Total for Question 4: 25 marks

## End of Model Paper 2

**BM3122 | MEDICAL IMAGING**

DOCUMENT B | MODEL 3

# Challenging application-based practice paper

University of Moratuwa-style undergraduate practice examination. Independently prepared; this is not an official university examination or a disclosure of the future paper.

## BM3122 MEDICAL IMAGING

### Time allowed: Two hours | Total: 100 marks

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• All constants and numerical conditions required are supplied. If interpreting a qualitative question, state your interpretation clearly.

• Suggested timing: 30 minutes per main question. Use separate answer sheets. Marking schemes are proposed and appear only in the solution documents.

**Candidate: _____ Date: _____**

Practice design: More demanding engineering applications within the supplied material.

# Information sheet | Model paper 3

This sheet is provided for these independent practice papers. Equations are corrected and assumptions are stated here; provision of a similar sheet in 2026 is not verified. Symbols have their usual meaning and units must be consistent.

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|---|---|
| Constants | $\gamma=267.52\times10^{6}$ rad $s^{-1}$ $T^{-1}$; $\gamma bar=\gamma/(2\pi)$. $h=6.63\times10^{-34}$ J s; $k=1.38\times10^{-23}$ J/K. Kelvin=°C+273.15. |
| Resolution and noise | FWHM of a Gaussian=$2\sqrt{(2\ln2)}\sigma$. I=O*h. $CNR=(S_A-S_B)/\sigma_N$. For independent noise $SNR_N=SNR_1\sqrt{N}$. ADC step=voltage span/$2^n$ for equal-width bins. |
| X-ray / CT | $I=I_0\exp(-\Sigma\mu_i x_i)$; $HVL=\ln2/\mu$. $p=-\ln(I/I_0)$. $HU=1000(\mu-\mu_{water})/\mu_{water}$. Water=0 HU; air≈-1000 HU. Effective focal spot f=F sinθ; coverage=2D tanθ. |
| Nuclear medicine | $A=-dN/dt=\lambda N$; $\lambda=\ln2/T_{half}$; $A(t)=A_0\exp(-\lambda t)$. $1/T_{eff}=1/T_{phys}+1/T_{bio}$ for independent exponential clearance. Non-paralyzable dead time: τ=(N-n)/(nN), where N,n are true/measured rates. |
| MRI encoding | $f_0=\gamma B_0/(2\pi)$. Full RF bandwidth $\Delta f=\gamma bar\,|G|\Delta z$. A central slice has $f_{min,max}=f_0\pm\Delta f/2$. $T_{scan}=TR\times N_{phase}\times NEX$ for conventional 2D acquisition. FOV=1/Δk. |
| MRI signal | Spoiled steady-state GE: $S=K\rho[(1-E_1)\sin\alpha/(1-E_1\cos\alpha)]\exp(-TE/T_2^*)$, $E_1=\exp(-TR/T_1)$. Ernst angle=$\arccos(E_1)$. Ideal spin echo: $S=K\rho[1-\exp(-TR/T_1)]\exp(-TE/T_2)$. Net population fraction=$\gamma hB_0/(4\pi kT)$. |
| Ultrasound | $Z=\rho c$. Intensity reflectance at normal incidence=$[(Z_2-Z_1)/(Z_2+Z_1)]^2$. λ=c/f. Round-trip propagation loss=2αfd, with d in cm, f in MHz and α in dB/(cm·MHz). Echo time=2d/c; frame rate≤$c/(2dN_{lines}N_{firings/line})$. Axial resolution=$N_{cycles}\lambda/2$; focal lateral resolution≈λF/L. |
| Optical / thermal / dose | Gaussian OCT axial resolution in air $\delta z=(2\ln2/\pi)\lambda_0^2/\Delta\lambda$; tissue $\delta z=\delta z_{air}/n$. Total thermal emission $M=\varepsilon\sigma T^4$. D=energy/mass (Gy); $H=w_R D$ (Sv); $E=\Sigma w_T H_T$ (Sv). Intensity loss from reflectance R: -10 $\log_{10}R$ dB. |

ART tasks use unit ray weights and additive residual / number-of-pixels updates as explicitly directed. Ultrasound timing assumes sequential lines unless stated otherwise. All models are answer-all papers. Useful approximate conversion: 1 mT/cm=0.1 T/m.

# MODEL 3 | Question 1 [25 marks]

## Image quality and quantitative nuclear imaging

**(a)** Two systems have normalized Gaussian PSFs. System A has σx=0.40 mm and σy=1.00 mm; B has σx=σy=0.70 mm. Calculate each directional FWHM and choose the better system for separating structures 1.2 mm apart along x, using separation greater than FWHM as the stated criterion. **[4]**

**(b)** Two regions have mean signals 120 and 100 units and noise standard deviation 5 units. Compute their CNR. A small lesion voxel contains 25% of the first region and 75% of the second. Find its mean signal and CNR against the second region, assuming unchanged noise. **[4]**

**(c)** An image with independent zero-mean noise has SNR=12. How many total independent acquisitions must be averaged to reach SNR=30? Explain why identical-view ultrasound speckle need not obey this scaling. **[2]**

**(d)** A tracer has physical half-life 8 h and biological half-life 12 h. Its initial in-body activity is 160 MBq. Calculate the effective half-life, activity after 6 h, and time at which activity first falls to 20 MBq. Explain the role of independent clearance. **[6]**

**(e)** A non-paralyzable gamma-camera channel records n=40,000 counts/s. Its dead time is τ=5.0 μs. Using τ=(N-n)/(nN), derive N in terms of n and τ, calculate the true rate N and the fraction of true events lost. **[5]**

**(f)** Describe true, random and scattered coincidences in PET. Explain one way that false coincidences affect the reconstructed image. **[4]**

Total for Question 1: 25 marks

# MODEL 3 | Question 2 [25 marks]

## Radiographic design and reconstruction constraints

**(a)** An X-ray target has actual projected electron-track width F=2.0 mm and bevel angle θ=12°. Calculate effective focal spot f=F sinθ and beam coverage 2D tanθ at D=80 cm. Recalculate at 8° and explain the image/FOV trade-off. **[4]**

**(b)** Without a grid, a detector receives 600 primary and 400 scattered photons in a stated interval. A grid transmits 70% of primary photons and 10% of scattered photons. Find the detected total and primary fraction with the grid. If output is scaled to restore the original total detected count, estimate the relative exposure factor; assume counts scale linearly with exposure. **[4]**

**(c)** A CT scanner measures air as -1040 HU and water as +20 HU. A linear calibration is h=a hm+b. Determine a and b using true air=-1000 HU and water=0 HU. Correct a measured tissue value of 232 HU, then compute its μ if μwater=0.200 cm^-1. **[5]**

**(d)** For a 2×2 image [[a,b],[c,d]], unit-weight projections are top row=8, bottom row=12, left column=7 and right column=13. Start from zero and use one equal-residual ART sweep in that order. Then an extra isolated ray gives a=2. Determine the unique image satisfying all data and verify it. Distinguish this final constrained solution from the one-sweep estimate. **[8]**

**(e)** Explain why noisy or incomplete projections can make exact ray-sum agreement undesirable or impossible. Give one advantage and one limitation of iterative reconstruction relative to direct analytic reconstruction. **[4]**

Total for Question 2: 25 marks

# MODEL 3 | Question 3 [25 marks]

## MRI signal optimization and acquisition design

**(a)** For a spoiled GE sequence, explain the roles of TR, TE, flip angle α and T2*. State which may be chosen by the operator and which is a tissue/field property. **[4]**

**(b)** For a fixed-TR, fixed-TE spoiled GE sequence with $E=\exp(-TR/T1)$, signal is proportional to $(1-E)\sin\alpha/(1-E\cos\alpha)$. Derive the maximizing angle for $0<\alpha<\pi/2$. Evaluate it for TR=0.40 s and T1=1.00 s. **[5]**

**(c)** At TR=0.40 s, TE=0.012 s and T1=1.00 s, compare two otherwise identical regions with T2*=0.040 s and 0.080 s. Use the Ernst angle from (b). Calculate their normalized GE signals for ρ=1, their signal ratio, and the percentage increase for the longer-T2* region. **[5]**

**(d)** A 256×128 readout-by-phase image uses TR=0.40 s and three averages. Calculate total scan time. If the phase steps are doubled and averages reduced to one, find the new scan time and explain the change. **[4]**

**(e)** Each complete slice sequence occupies 25 ms within TR=0.40 s. Find the maximum ideal interleaved slice count and the scan time for 12 slices using the original settings of (d). Explain the required frequency and storage changes. **[4]**

**(f)** Under the approximate scaling SNR proportional to voxel volume times √(total ADC-on time), what happens to SNR if voxel volume is halved with unchanged ADC-on time? How much additional acquisition time would restore the original SNR? State a practical cost. **[3]**

Total for Question 3: 25 marks

# MODEL 3 | Question 4 [25 marks]

## Ultrasound design and optical ranging

**(a)** A target reflects 50% of incident intensity and lies in tissue with attenuation 0.60 dB/(cm·MHz). At f=3 MHz the total permissible echo-intensity loss is 90 dB, measured relative to launched intensity. Calculate the maximum depth, accounting for reflection and round-trip propagation. Ignore all other losses. **[6]**

**(b)** At 3 MHz and c=1540 m/s, the pulse contains three cycles. The focal distance is 60 mm and aperture width 20 mm. Compute wavelength, axial resolution and focal lateral resolution using the supplied lecture approximations. What would doubling frequency do if all other values remained fixed? **[5]**

**(c)** A 128-line frame images to 20 cm at c=1540 m/s. Compare maximum frame rates for one transmit focus per line and three sequential transmit foci per line. Assume no parallel reception or timing overhead. **[4]**

**(d)** Explain angular and frequency compounding for speckle reduction. State why improved smoothness can conceal a loss of temporal resolution. **[4]**

**(e)** An OCT source has Gaussian spectrum with centre wavelength $\lambda_0$=840 nm and FWHM bandwidth $\Delta\lambda$=60 nm. Using $\delta z_{air}=(2\ln 2/\pi)\lambda_0^2/\Delta\lambda$, find axial resolution in air and in tissue of refractive index n=1.40, taking $\delta z_{tissue}=\delta z_{air}/n$. Explain why a broader spectrum improves axial resolution and why this calculation does not determine lateral resolution. **[6]**

Total for Question 4: 25 marks

## End of Model Paper 3

DOCUMENT B | MODEL 4

# Full-syllabus coverage practice paper

University of Moratuwa-style undergraduate practice examination. Independently prepared; this is not an official university examination or a disclosure of the future paper.

## BM3122 MEDICAL IMAGING

### Time allowed: Two hours | Total: 100 marks

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• All constants and numerical conditions required are supplied. If interpreting a qualitative question, state your interpretation clearly.

• Suggested timing: 30 minutes per main question. Use separate answer sheets. Marking schemes are proposed and appear only in the solution documents.

**Candidate: _____ Date: _____**

Practice design: Broad syllabus revision; several assignments and syllabus topics have limited direct past-paper precedent.

# Information sheet | Model paper 4

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| Resolution and noise | FWHM of a Gaussian$=2\sqrt{(2\ln2)}\sigma$. I=O*h. $CNR=(S_A-S_B)/\sigma_N$. For independent noise $SNR_N=SNR_1\sqrt{N}$. ADC step=voltage span/$2^n$ for equal-width bins. |
| X-ray / CT | $I=I_0\exp(-\Sigma\mu_i x_i)$; $HVL=\ln2/\mu$. $p=-\ln(I/I_0)$. $HU=1000(\mu-\mu_{water})/\mu_{water}$. Water=0 HU; air≈-1000 HU. Effective focal spot f=F sinθ; coverage=2D tanθ. |
| Nuclear medicine | $A=-dN/dt=\lambda N$; $\lambda=\ln2/T_{half}$; $A(t)=A_0\exp(-\lambda t)$. $1/T_{eff}=1/T_{phys}+1/T_{bio}$ for independent exponential clearance. Non-paralyzable dead time: $\tau=(N-n)/(nN)$, where N,n are true/measured rates. |
| MRI encoding | $f_0=\gamma B_0/(2\pi)$. Full RF bandwidth $\Delta f=\gamma bar\,\lvert G\rvert\Delta z$. A central slice has $f_{min,max}=f_0\pm\Delta f/2$. $T_{scan}=TR\times N_{phase}\times NEX$ for conventional 2D acquisition. $FOV=1/\Delta k$. |
| MRI signal | Spoiled steady-state GE: $S=K\rho[(1-E_1)\sin\alpha/(1-E_1\cos\alpha)]\exp(-TE/T_2^*)$, $E_1=\exp(-TR/T_1)$. Ernst angle=$\arccos(E_1)$. Ideal spin echo: $S=K\rho[1-\exp(-TR/T_1)]\exp(-TE/T_2)$. Net population fraction=$\gamma hB_0/(4\pi kT)$. |
| Ultrasound | $Z=\rho c$. Intensity reflectance at normal incidence=$[(Z_2-Z_1)/(Z_2+Z_1)]^2$. $\lambda=c/f$. Round-trip propagation loss=$2\alpha fd$, with d in cm, f in MHz and α in dB/(cm·MHz). Echo time=2d/c; frame rate≤$c/(2dN_{lines}N_{firings/line})$. Axial resolution=$N_{cycles}\lambda/2$; focal lateral resolution≈λF/L. |
| Optical / thermal / dose | Gaussian OCT axial resolution in air $\delta z=(2\ln2/\pi)\lambda_0^2/\Delta\lambda$; tissue $\delta z=\delta z_{air}/n$. Total thermal emission $M=\varepsilon\sigma T^4$. D=energy/mass (Gy); $H=w_R D$ (Sv); $E=\Sigma w_T H_T$ (Sv). Intensity loss from reflectance R: -10 $\log_{10}R$ dB. |

ART tasks use unit ray weights and additive residual / number-of-pixels updates as explicitly directed. Ultrasound timing assumes sequential lines unless stated otherwise. All models are answer-all papers. Useful approximate conversion: 1 mT/cm=0.1 T/m.

# MODEL 4 | Question 1 [25 marks]

## Sampling, image processing and nuclear instrumentation

**(a)** An ideal 10-bit rounding ADC has input range 0-2.048 V and uses 1024 equal-width bins. Calculate its code width and maximum quantization-error magnitude. A useful input signal spans only 0-0.128 V; propose a gain to use the full range and explain the risk. Give three ADC specifications relevant to imaging. **[5]**

**(b)** A real low-pass analogue signal contains frequencies up to 18 kHz. State the theoretical Nyquist sampling bound and explain the role of an analogue anti-alias filter. A 16-fold oversampling/decimation scheme is then used; what ideal equivalent-bit improvement is suggested by the lecture, and under what noise assumption? **[4]**

**(c)** Compare spatial low-pass and high-pass filtering in terms of detail, noise and diagnostic visibility. **[3]**

**(d)** Draw the signal path of a scintillation gamma camera. Explain the roles of collimator, scintillator, photodetectors, position/energy electronics and energy-window selection. **[5]**

**(e)** Explain why SPECT needs scatter and attenuation correction. State how a combined SPECT/CT system assists quantitative interpretation. **[4]**

**(f)** Using a simple sketch, distinguish a PET line of response from the position of an annihilation event. Explain why the detection of one coincidence does not by itself create a complete image. **[4]**

Total for Question 1: 25 marks

# MODEL 4 | Question 2 [25 marks]

## Radiography and CT beyond the common repeats

**(a)** Explain the origin of the continuous and characteristic components of an X-ray-tube spectrum. State the different primary effects of increasing kVp and increasing mAs. **[4]**

**(b)** Compare photoelectric absorption and Compton scattering for diagnostic X-rays. Explain why pair production is not significant at a maximum tube photon energy of 120 keV. **[4]**

**(c)** Compare computed radiography (CR), indirect digital radiography and direct digital radiography by their signal-conversion and readout principles. **[4]**

**(d)** A mass of 0.20 kg absorbs 0.60 mJ of X-ray energy. Calculate absorbed dose in Gy. With radiation weighting factor wR=1, determine equivalent dose in Sv. If only this tissue is exposed and its tissue weighting factor is 0.12, calculate its contribution to effective dose. Distinguish these three quantities. **[5]**

**(e)** For this question define multi-detector helical pitch as table travel per rotation divided by total active collimated beam width. A scanner has 16 active rows each 1.25 mm wide, table travel 30 mm/rotation and rotation time 0.50 s. Calculate pitch and table speed; neglect end effects and find time to cover 300 mm. Explain why the pitch convention must be stated. **[4]**

**(f)** Explain the causes of (i) a ring artifact in third-generation CT and (ii) cupping in a uniform phantom. Give one corrective action for each. **[4]**

Total for Question 2: 25 marks

# MODEL 4 | Question 3 [25 marks]

## MRI relaxation, reconstruction and functional imaging

**(a)** After a 90° excitation, write and sketch ideal longitudinal recovery and transverse decay. Distinguish T1, T2 and T2*. **[4]**

**(b)** Explain how phase- and frequency-encoding gradients sample a 2D k-space grid, how the image is reconstructed, and how reducing $\Delta k$ or increasing the maximum sampled spatial frequency changes the image. **[5]**

**(c)** State suitable qualitative TR and TE choices for proton-density-weighted, T1-weighted and T2-weighted conventional spin-echo images. Explain why long TR reduces T1 contrast. **[4]**

**(d)** An object extends beyond the encoded field of view in the phase direction. Explain wraparound aliasing and give two acquisition remedies with their trade-offs. **[3]**

**(e)** Explain the BOLD basis of fMRI, why rapid EPI acquisition is useful, and how a simple repeated task/rest experiment identifies associated regions. State one interpretation limitation. **[5]**

**(f)** Explain time-of-flight MR angiography in terms of inflowing blood and saturated stationary tissue. Why can a large tip angle relative to the stationary tissue Ernst angle improve vessel contrast, and what is one limitation? **[4]**

Total for Question 3: 25 marks

# MODEL 4 | Question 4 [25 marks]

## Ultrasound instrumentation, OCT and thermography

**(a)** Explain piezoelectric transmission and reception in an ultrasound probe. Sketch a basic pulse-echo system signal path including a transmitter/receiver switch and display. **[4]**

**(b)** Compare linear and phased arrays for beam formation. Explain how electronic focusing improves lateral resolution and give one cost of using several transmit focal zones. **[4]**

**(c)** Explain what Doppler ultrasound adds to B-mode imaging. State how motion towards versus away from the probe affects the detected frequency, and why perpendicular flow is difficult to measure with this principle. **[3]**

**(d)** State two physical reasons why ultrasound may fail to provide a useful image of a target behind an air-filled cavity or thick bone. **[2]**

**(e)** Compare time-domain and Fourier-domain OCT. Explain A-scan-to-B-scan formation, give one ophthalmic application, and identify one acquisition or processing error that could corrupt a thickness measurement. **[6]**

**(f)** A thermal camera compares a body surface at 307 K with a reference surface at 305 K. Assuming equal emissivity and the supplied total-emission law M=εσT^4, calculate the percentage increase of emitted power. Explain why the change is not by itself a diagnosis, and give three controls for a reliable comparison. **[6]**

Total for Question 4: 25 marks

## End of Model Paper 4